Nanyi Zheng

Postdoctoral Researcher in Applied and Computational Mathematics

Department of Mathematical Sciences, University of Delaware, Newark, DE, USA

nyzheng@udel.edu

Research Interests

Numerical methods for kinetic equations and partial differential equations; adaptive low-rank and tensor methods for high-dimensional problems; structure-preserving semi-Lagrangian methods; high-order WENO and HWENO methods for hyperbolic conservation laws.

Academic Appointment

University of Delaware, Department of Mathematical Sciences **Jul. 2023–present**

Postdoctoral Researcher; advisor: Prof. Jing-Mei Qiu

Education

Xiamen University **Sep. 2018–Jun. 2023**

Ph.D. in Computational Mathematics; advisor: Prof. Jianxian Qiu

Dissertation: High-Order Semi-Lagrangian Finite Volume WENO-Type Methods for Transport Equations

Xiamen University **Sep. 2014–Jun. 2018**

B.S. in Information and Computing Science

Publications

Nanyi Zheng is shown in bold; † denotes corresponding author.

1. **Nanyi Zheng**, William A. Sands, Daniel Hayes, Andrew J. Christlieb, and Jing-Mei Qiu. "A Semi-Lagrangian Adaptive Rank (SLAR) Method for High-Dimensional Vlasov Dynamics." *SIAM Journal on Scientific Computing*, accepted, Jul. 1, 2026. arXiv:2510.24861.
2. Andrew J. Christlieb, Sining Gong, Jing-Mei Qiu, and **Nanyi Zheng**. "A Sampling-Based Adaptive-Rank Approach to the Wigner–Poisson System." *SIAM Journal on Scientific Computing*, 48(4), B698–B723, 2026. doi:10.1137/25M1776019.
3. William A. Sands, Jing-Mei Qiu, Daniel Hayes, and **Nanyi Zheng**†. "An Adaptive-Rank Approach with Greedy Sampling for Multi-Scale BGK Equations." *Journal of Computational Physics*, 547, 114523, 2026. doi:10.1016/j.jcp.2025.114523.
4. **Nanyi Zheng**, Daniel Hayes, Andrew J. Christlieb, and Jing-Mei Qiu. "A Semi-Lagrangian Adaptive-Rank (SLAR) Method for Linear Advection and Nonlinear Vlasov–Poisson System." *Journal of Computational Physics*, 532, 113970, 2025. doi:10.1016/j.jcp.2025.113970.
5. **Nanyi Zheng**, Xiaofeng Cai, Jing-Mei Qiu, and Jianxian Qiu. "Non-Splitting Eulerian–Lagrangian WENO Schemes for Two-Dimensional Nonlinear Convection–Diffusion Equations." *Journal of Computational Physics*, 529, 113890, 2025. doi:10.1016/j.jcp.2025.113890.
6. **Nanyi Zheng**, Xiaofeng Cai, Jing-Mei Qiu, and Jianxian Qiu. "Fourth-Order Conservative Non-Splitting Semi-Lagrangian Hermite WENO Schemes for Kinetic and Fluid Simulations." *Journal of Scientific Computing*, 99(3), Article 70, 2024. doi:10.1007/s10915-024-02520-6.
7. **Nanyi Zheng**, Xiaofeng Cai, Jing-Mei Qiu, and Jianxian Qiu. "A Fourth-Order Conservative Semi-Lagrangian Finite Volume WENO Scheme without Operator Splitting for Kinetic and Fluid Simulations." *Computer Methods in Applied Mechanics and Engineering*, 395, 114973, 2022. doi:10.1016/j.cma.2022.114973.
8. **Nanyi Zheng**, Xiaofeng Cai, Jing-Mei Qiu, and Jianxian Qiu. "A Conservative Semi-Lagrangian Hybrid Hermite WENO Scheme for Linear Transport Equations and the Nonlinear Vlasov–Poisson System." *SIAM Journal on Scientific Computing*, 43(5), A3580–A3606, 2021. doi:10.1137/20M1363273.

Manuscripts Under Review

1. **Nanyi Zheng**, William A. Sands, and Jing-Mei Qiu. "A Mass, Momentum, and Energy Conserving Semi-Lagrangian Adaptive-Rank (SLAR) Method for the Vlasov–Poisson System." Submitted to *SIAM Journal on Scientific Computing*, 2026. arXiv:2606.29027.
2. Andrew J. Christlieb, Sining Gong, Jing-Mei Qiu, and **Nanyi Zheng**. "A Structure-Preserving Adaptive-Rank Approach to the High-Dimensional Wigner–Poisson System." Submitted to *SIAM Journal on Scientific Computing*, 2026. arXiv:2606.15067.

Open-Source Research Software

- **High-Dimensional SLAR** (MATLAB, 2025): adaptive-rank Vlasov–Poisson solvers for 2D2V and 3D3V configurations (four- and six-dimensional phase spaces), including Hierarchical Tucker Adaptive Cross Approximation routines.
- **LoMaC-SLAR** (MATLAB, 2025): locally macroscopic conservative SLAR solvers for Vlasov–Poisson simulations in 1D1V and 2D2V configurations, designed to preserve mass, momentum, and energy.

Funded Research Experience

Tensor Approaches for Simulating Kinetic Systems (TASKS)
AFOSR MURI; overall award: \$7.5M; University of Delaware PI: Prof. Jing-Mei Qiu **Jul. 2025–present**

- Postdoctoral researcher on the University of Delaware team, contributing to algorithm design, mathematical derivation, software implementation, and numerical validation for high-dimensional kinetic solvers.

A Conservative Low Rank High Order Tensor Approach for Nonlinear Vlasov Equations
AFOSR basic research project; award: \$425,506; PI: Prof. Jing-Mei Qiu **Jul. 2023–Jun. 2025**

- Project-supported postdoctoral researcher responsible for numerical method development, mathematical derivation, research software, and computational verification.

Teaching Experience

University of Delaware, Department of Mathematical Sciences **Fall 2023–present**
Instructor

- *Differential Equations with Linear Algebra II*: Fall 2023, Spring 2024, Fall 2024, Spring 2025, Fall 2025, and Fall 2026 (current).
- *Computational Mathematics II*: Spring 2026.

Invited and Contributed Talks

- **Invited talk**, 28th International Conference on Numerical Simulation of Plasmas (ICNSP), 2025. Opening 45-minute presentation: "A Semi-Lagrangian Adaptive Rank (SLAR) Method for High-Dimensional Vlasov Dynamics."
- **Contributed talk**, SIAM New York–New Jersey–Pennsylvania Section Meeting (SIAM-NNP), 2024. "Semi-Lagrangian Adaptive-Rank Methods for Linear Advection Equations and the Nonlinear Vlasov–Poisson System."

Professional Service

- Ad hoc referee for *SIAM Journal on Scientific Computing* (2026), *Journal of Computational Physics* (2024, 2026), and *Advances in Computational Science and Engineering* (2026).